

Shaping HCI Research for Children's Care Ecosystem Involvement

Evropi Stefanidi
TU Wien
Vienna, Austria
evropi.stefanidi@tuwien.ac.at

Lucas M. Silva
Computer Science
University of Iowa
Iowa City, Iowa, USA
lucas-silva@uiowa.edu

Bengisu Cagiltay
Koç University
Istanbul, Türkiye
bcagiltay@ku.edu.tr

Aehong Min
Department of Informatics
University of California, Irvine
Irvine, California, USA
ahmin912@gmail.com

Eva Eriksson
Aarhus University
Aarhus, Denmark
evae@cc.au.dk

Gillian R Hayes
Department of Informatics
University of California, Irvine
Irvine, California, USA
gillianrh@ics.uci.edu

Abstract

Recent HCI research emphasizes the importance of considering children's care ecosystems in the design of technologies, extending the focus beyond families to include teachers, peers, therapists, and institutions. While this ecosystem perspective opens opportunities for more inclusive and collaborative technologies, it also introduces challenges such as recruitment, power dynamics, reconciling diverse perspectives, and complex ethical considerations. This CHI 2026 workshop builds on prior community efforts at IDC 2023, CHI 2024, and IDC 2025. Its primary focus is on children's care ecosystems, but we also welcome researchers working with other populations who wish to apply an ecosystem lens. The workshop will bring together researchers and practitioners to discuss opportunities, challenges, and methods, and to collaboratively articulate a research agenda for care ecosystem-centered HCI. Participants will engage in mapping and synthesis activities that produce care ecosystem maps and a shared artifact capturing priorities and open questions for care ecosystem-centered HCI, while also fostering collaboration and building community ties across disciplines and stakeholder contexts.

CCS Concepts

• Human-centered computing → User centered design.

Keywords

child-computer interaction, children, families, care ecosystem, caregiver, multi-stakeholder

ACM Reference Format:

Evropi Stefanidi, Lucas M. Silva, Bengisu Cagiltay, Aehong Min, Eva Eriksson, and Gillian R Hayes. 2026. Shaping HCI Research for Children's Care Ecosystem Involvement. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '26, Barcelona, Spain

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3778689>

Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3778689>

1 Motivation

Children's development and their engagement with technology are situated in care ecosystems that include parents, siblings, peers, teachers, therapists, and broader community and institutional actors [1]. Recent HCI work has begun to highlight the importance of engaging with these broader ecosystems in order to design technologies that better support children's well-being (e.g., [4, 5, 7, 10]). Over the past three years, a series of workshops has shown a growing recognition that children's interactions with technology are embedded in complex social contexts that extend beyond the child alone. At IDC 2023, the community turned its attention from child-centered to family-centered approaches [3], highlighting the importance of considering family members' roles and dynamics. Building on this foundation, the CHI 2024 workshop [2] focused on methodological strategies for engaging children and parents together, exploring the benefits and tensions of family-centered design. Our IDC 2025 workshop [11] took the next step by broadening the perspective to care ecosystems—networks that include not only families but also teachers, peers, therapists, clinicians, and institutional actors who shape children's daily experiences. Related themes have also been explored at CHI, such as a 2022 workshop on complex health ecosystems [8]. While that work emphasized clinical infrastructures, our workshop focuses on children's everyday care ecosystems across contexts. Thus, our proposed workshop continues this progression from both CHI and IDC workshops by consolidating insights from these earlier efforts and producing a tangible *Care Ecosystem Research Agenda* that outlines priorities, challenges, and future directions for HCI.

Designing with care ecosystems offers clear opportunities. By engaging multiple stakeholders, technologies might better integrate within children's lived realities and support their well-being across home, school, clinical, and community contexts [9]. Yet these opportunities come with considerable challenges for research and design. Recruitment becomes more demanding as researchers must reach contexts and stakeholders beyond families, all of whom operate under different constraints [6]. Involving multiple roles and

stakeholders can also incur complex system and interaction requirements, which might integrate care but place substantial burdens on participants (e.g., device management, system workflow, multi-step and multi-user tasks). Multi-stakeholder design commitment can act as a barrier to participation, especially for families managing complex care arrangements. Researchers must therefore invest significant effort into study planning, system designs, and maintaining sustainable relationships with participants across different roles [13].

Integrating collaboration within care ecosystems can face challenges with balancing diverse perspectives and responsibilities. Children, parents, educators, and professionals often hold divergent values and expectations, and these may come into conflict. For example, in our MoodGems project [12], children and therapists welcomed tools that gave them control over personal data, while some parents opposed that, wishing for full access to their children's data. Such tensions underscore the difficulty of balancing agency, trust, and oversight in care ecosystems. They also raise broader methodological and ethical questions: how should consent be designed when multiple parties are involved? How can power asymmetries between children and adults, or between families and institutions, be acknowledged and addressed? Should stakeholders be engaged together in shared sessions, or separately to avoid dominance effects? Should systems adapt interactions to each role, or provide uniform experiences? Each of these questions points to trade-offs that matter both for research validity and for the adoption of technologies in practice.

While our primary focus is on children's care ecosystems, we recognize that the ecosystem perspective has wider relevance. Researchers working with other populations (e.g., homebound older adults or adults with motor impairments) may also wish to explore how an ecosystem lens can illuminate multi-stakeholder dynamics. This workshop, therefore, offers a space that centers children while remaining open to broader applications of ecosystem-centered design in HCI.

These research and design challenges highlight why a dedicated space for consolidating knowledge and setting directions is needed. This CHI 2026 workshop creates such a space by bringing together researchers and practitioners to reflect on prior experiences, share methodological innovations, and collectively identify open questions around centering HCI on care ecosystem support. Our primary goal is to draft a research agenda that defines pressing challenges and methodological opportunities for care ecosystem-centered HCI. Alongside this agenda, the workshop will foster peer feedback on ongoing projects, surface strategies for multi-stakeholder engagement, and strengthen the community of researchers working in this space. Through these efforts, we aim not only to extend the trajectory initiated at IDC 2023, CHI 2024, and IDC 2025, but also to produce a tangible artifact, in the form of a Care Ecosystem Research Agenda, capturing the community's priorities, unresolved challenges, and future directions. At the same time, this workshop aims to build enduring connections that advance research on children's broader support networks.

2 Length of the Workshop

One 90-minute, in-person session at CHI 2026.

3 Organizers

Evropi Stefanidi (Main Contact), PhD, is a postdoctoral researcher at the HCI Group of TU Wien. Her research focuses on designing technologies for and with children and their care ecosystems. She has experience in conducting multi-stakeholder research with both younger children and teenagers, and various care ecosystem members such as parents, siblings, grandparents, peers, teachers, and therapists. Together with other organizers of this workshop, she hosted the care ecosystems workshop at IDC 2025.

Lucas M. Silva, PhD, is an Assistant Professor in Computer Science at The University of Iowa, USA. His research focuses on investigating the design of systems that help people discover, understand, and interact with personal data to improve health and well-being, both individually and in collaboration within care ecosystems. Many of his projects are in partnership with neurodivergent children and involve technology-mediated co-regulation. He was involved in organizing the IDC 2025 care ecosystem workshop.

Bengisu Çağıltay, PhD, is an incoming Assistant Professor of Media and Visual Arts at Koç University, Turkey. Her research interests broadly focus on designing socially interactive robots to support children and families' everyday lives. She has hosted similar workshops on the topic of family-centered design at IDC 2025, IDC 2023, and CHI 2024.

Aehong Min, PhD, is a postdoctoral scholar in Informatics at the University of California, Irvine. Her research focuses on informal caregivers and technologies that support their collaborative work with others, with the goal of reducing caregiving burdens and promoting the health and well-being of all stakeholders within the care network. She was part of the organizers of the complex healthcare ecosystems workshop at CHI 2022.

Eva Eriksson, PhD, is an Associate Professor in Interaction Design at Aarhus University in Denmark. Eva specializes in developing technologies through participatory design in the field of child-computer interaction with a focus on public learning institutions. She has led several research projects focusing on scaffolding children's collaboration mediated by technologies in special education contexts, and was part of the workshop organization team of the IDC 2025 care ecosystems workshop.

Gillian R. Hayes, PhD, is the Chancellor's Professor and the Robert A. and Barbara L. Kleist Professor of Informatics at the University of California, Irvine. Her research focuses on social, political, and technological interventions for human needs in sensitive and ethically responsible ways.

4 Pre-Workshop Plans

Here, we outline our pre-workshop plans for recruiting participants and community building.

Recruitment. We will invite researchers and practitioners working with children's broader support networks (families, educators, therapists, clinicians, community organizations) via academic mailing lists (e.g., CHI, CHI-Kids), and social media (e.g., BlueSky, LinkedIn). Participation will be possible via either (i) a short statement of interest (~250 words) outlining a project or challenge, or (ii) a 2–4 page position paper aligned with the themes in Section 10. Submissions will be lightly reviewed for relevance and diversity of perspectives.

Website. We will host a Google Sites page (<https://sites.google.com/view/chi26-ecocare/home>) with the call for participation, objectives, organizer contacts, key dates, and a tentative program. Prior to CHI, we will also make participants' submissions available on the website.

Pre-Workshop Materials. To facilitate interdisciplinary participation within the limited time of the workshop, we will already share relevant materials with participants before the workshop, including Bronfenbrenner's ecological systems theory [1], given its central role in the workshop activities and conceptualisation.

Community Building. One week before CHI, participants will be invited to a Discord space, where they will have the opportunity to introduce themselves and network. Following the workshop, the space will remain open for sharing updates and exploring collaboration opportunities.

5 Plans to Publish Workshop Proceedings

Our main outcome will be a *Care Ecosystem Research Agenda* that synthesizes clustered challenges and open questions across ecological levels. We will circulate a draft to participants for feedback and invite them to co-author or endorse the final version. The finalized artifact will be shared openly on the workshop website. If we receive a substantial number of position papers, we will also publish them on our website in a single PDF compilation.

6 Accessibility

We will ensure that the workshop is accessible to all participants. If needed, we will use a live captioning service for participants to follow discussions in real time. All materials will be available as screen-reader-friendly PDFs, and participants will be welcome to contribute verbally or in writing during activities.

7 Offline Materials

All workshop materials, including the slide deck, the care ecosystem maps, and the final research agenda artifact, will be available as downloadable PDFs after the session. We will also share photographs or digital exports of the completed maps (with consent) and a short summary of key takeaways for those who could not attend.

8 Workshop Activities

The 90-minute session is structured as a progression from framing and group mapping to peer feedback and collective synthesis. Each stage contributes directly to our primary goal: the co-creation of a *research agenda for care ecosystem-centered HCI*. Along the way, the activities foster opportunities for peer exchange, methodological reflection, and community building. Table 1 provides an overview of the workshop schedule.

Welcome and Framing (15 minutes). We will begin by welcoming participants and situating this workshop within the broader trajectory of prior community efforts, moving from IDC 2023 [3] (child–family) to CHI 2024 [2] (family-centered) and IDC 2025 [11] (care ecosystems). We will then introduce Bronfenbrenner's ecological systems theory [1] as the conceptual scaffold for the session, presenting a simple visual to illustrate the micro, meso, exo, and

macro levels. To create an open and collaborative atmosphere, we will begin by running a short ice-breaker activity.

Care Ecosystem Mapping (25 minutes). We will group participants by thematic focus, informed by their submissions or statements of interest (e.g., methodological approaches, ethical considerations, or design contexts such as home, school, or clinical settings). Each group will be assigned a case to work on, ideally contributed by a participant, but otherwise provided as a seed case prepared by the organizers (for example, a wearable device connecting home and therapy for ADHD, or a classroom robot designed to bridge school and family contexts). Within their groups, participants will first **individually map** an ecosystem from their own perspective, based on their own research or the seed cases we will provide to all groups. To support this activity, organizers will provide ready-made mapping templates structured by Bronfenbrenner's ecological levels. Groups will then share these sketches and agree on a single case to develop further, using the templates as scaffolds, so that they **collaboratively create a care ecosystem map** that identifies relevant stakeholders across the ecological levels, articulates their goals and values, and highlights potential tensions or blind spots. Groups will be encouraged to cover all four levels of Bronfenbrenner's model, while focusing more deeply on those most central to their chosen case. Prompts will guide discussion on stakeholder relationships, conflicts such as child autonomy versus parental oversight, and perspectives that are often excluded from design. Each group will produce a visual map (on paper) that captures stakeholders, relationships, and tensions within their ecosystem.

Peer Feedback and Expansion (20 minutes). To enrich the maps with additional perspectives, groups will rotate to review one another's work. Some participants will remain with their group's map to provide a short overview when visitors arrive, while the rest of the group will visit another table. Visiting participants will annotate the map with feedback using structured prompts that ask, for example, which stakeholders or levels appear underrepresented, which perspectives might be revealed by different methods, or what new tensions emerge when zooming one level in or out. Groups will then return to their own map to review and integrate the comments. This stage ensures that each artifact incorporates both the reasoning of its creators and the constructive critique of peers.

Research Agenda Building (20 minutes). We will reconvene in plenary to synthesize the insights generated in the group work. Each group will briefly share one or two challenges from their case that they see as broadly relevant. An organizer will record these contributions on a central board, which will serve as the *Care Ecosystem Research Agenda* wall. Together with the participants, we will cluster related points into 4–6 broad themes, such as recruitment, multi-party consent, sustaining engagement, or institutional barriers. If time permits, we will also collaboratively formulate one or two open research questions for each theme and position them under the relevant ecological level (micro, meso, exo, or macro). If time runs short, the clustering and question formulation will be completed by the organizers after the session and shared back with participants for validation. The Research Agenda thus becomes the central output of the workshop, providing a tangible, collaboratively authored artifact that captures the community's priorities.

Table 1: Workshop Schedule Overview

Activity	Time	Summary
Welcome and Framing	15 min	Introduce the workshop trajectory and Bronfenbrenner’s model, set expectations, and run a short ice-breaker.
Care Ecosystem Mapping	25 min	Groups first sketch individually and then collaboratively create a care ecosystem map based on participant or seed cases, guided by Bronfenbrenner’s levels.
Peer Feedback and Expansion	20 min	Groups rotate to review each other’s maps, annotate them with structured feedback, and integrate new perspectives.
Research Agenda Building	20 min	Plenary synthesis of group insights, clustering into broad themes and formulating open research questions to create the Care Ecosystem Research Agenda (our main outcome artifact).
Wrap-Up and Commitments	10 min	Participants share takeaways and next steps; organizers summarize outcomes and invite continued collaboration via Discord.

Wrap-Up and Commitments (10 minutes). In the final stage, we will reflect on the day’s outcomes and invite participants to articulate personal takeaways. Each participant will be asked to share one key insight and one concrete next step for their own research. The organizers will summarize what has so far been collected on the Research Agenda, explain how these will be consolidated into a post-workshop artifact, and invite participants to contribute as co-authors. The session will close with an invitation to continue networking and collaboration via the dedicated Discord channel.

Expected Outputs. By the end of the workshop, participants will have (i) care ecosystem maps enriched by peer feedback, (ii) exposure to practical methods for navigating multi-stakeholder complexity, (iii) explicit takeaways and next steps for their own projects, and (iv) a collaboratively drafted *Care Ecosystem Research Agenda* that captures the community’s priorities and unresolved questions for future work.

9 Post-Workshop Plans

Here, we detail our post-workshop plans, including the dissemination of the research agenda artifact, the post-workshop report, and community building beyond the workshop.

Research Agenda Artifact. The central outcome of the workshop will be the collaboratively created *Care Ecosystem Research Agenda*. After the workshop, the organizers will consolidate the information available on the wall, together with notes from group maps and plenary clustering, into a written artifact. This document will summarize cross-cutting challenges, methodological opportunities, and open research questions, structured by ecological levels (micro, meso, exo, macro). We will circulate a draft to participants for feedback and invite them to co-author or endorse the final version. Afterwards, the artifact will be shared both on Discord and openly through the workshop website, and will be deposited in a persistent repository (e.g., arXiv), ensuring citable and long-term accessibility.

Workshop Report. Alongside the agenda, we will prepare a short report that documents the workshop process, showcases examples of ecosystem maps (with participant consent), and highlights participants’ reflections and commitments. This report will serve as a record of the workshop and a practical reference for others designing multi-stakeholder studies. The workshop report will be made available on the website, alongside the Research Agenda Artifact.

Community Building Beyond the Workshop. We aim to maintain the momentum of the workshop beyond CHI. The Discord space will remain active as a hub for sharing methods, calls, and collaboration opportunities. Participants will be encouraged to post updates on projects that connect to the research agenda, enabling ongoing knowledge exchange. We also plan to host a virtual follow-up session within two months of CHI to revisit the agenda, identify emerging collaborations, and discuss potential joint outputs such as panel proposals, cross-institutional studies, or special issue initiatives.

10 Call for Participation

We invite researchers, designers, and practitioners to join our 90-minute, in-person CHI 2026 workshop on designing technologies for and with children and their broader care ecosystems. While the primary focus is on children’s care ecosystems, we also welcome participants who are interested in applying an ecosystem lens to other populations. We invite contributions that address theoretical, methodological, or practical aspects of care ecosystem-centered design across home, school, clinical, and community settings. Participants may signal interest by either: (i) submitting a brief paragraph (~250 words) outlining their background and an ecosystem-related project or challenge, **or** (ii) submitting a 2–4 page position paper (including references) addressing: (a) ongoing or planned HCI work with children and their care ecosystems (e.g., families, educators, therapists, and other stakeholders), and/or (b) challenges and design practices for multi-stakeholder research, including open questions to be discussed that can shape the research agenda. Please submit PDFs via email to the organizers. At least one author of each accepted contribution must register for and attend both the workshop and the main conference, and all workshop participants must register for both. We look forward to engaging discussions on advancing care ecosystem-centered design at CHI 2026!

11 Expected Attendance

We target approximately 15 participants to enable deep, hands-on collaboration. Selection aims for diversity across roles (academia / industry / clinicians / educators), experience levels, and topical areas. If oversubscribed, we will prioritize contributions that bring distinct stakeholder perspectives or contexts (e.g., school, clinic, community) to maximize cross-pollination during group work.

12 Note About Past Workshops

This workshop extends a sequence of community efforts moving from child-centered to family-centered and, most recently, care ecosystem-centered design. It builds on the IDC 2023 workshop on family-centered interaction design [3], the CHI 2024 workshop on methods for family-centered design [2], and our IDC 2025 workshop focused explicitly on children's care ecosystems [11]. The CHI 2026 workshop consolidates insights from this series to articulate a forward-looking research agenda for care ecosystem-centered HCI.

Acknowledgments

This work was supported in part by the Independent Research Fund Denmark (nr. 3167-00011B) in the COMPILE project.

References

- [1] Urie Bronfenbrenner and Pamela A Morris. 2007. The Bioecological Model of Human Development. *Handbook of Child Psychology* (jun 2007). doi:10.1002/9780470147658.CHPSY0114
- [2] Bengisu Cagiltay, Hui-Ru Ho, Kaiwen Sun, Zhaoyuan Su, Yuxing Wu, Olivia K Richards, Qiao Jin, Junnan Yu, Jerry Alan Fails, Jason Yip, et al. 2024. Methods for Family-Centered Design: Bridging the Gap Between Research and Practice. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, 1–6. doi:10.1145/3613905.3636290
- [3] Bengisu Cagiltay, Rabia Ibtasar, Joseph E Michaelis, Sarah Sebo, and Bilge Mutlu. 2023. From Child-Centered to Family-Centered Interaction Design. In *Proceedings of the 22nd Annual ACM Interaction Design and Children Conference*. Association for Computing Machinery, 789–791. doi:10.1145/3585088.3589930
- [4] Franceli L Cibrian, Kimberley D Lakes, Arya Tavakoulnia, Kayla Guzman, Sabrina Schuck, and Gillian R Hayes. 2020. Supporting self-regulation of children with ADHD using wearables: tensions and design challenges. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–13.
- [5] Robert D Kalinowski, Ying Xu, and Katie Salen. 2021. The Ecological Context of Preschool-Aged Children's Selection of Media Content. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 349, 14 pages. doi:10.1145/3411764.3445429
- [6] Jonathan Lazar, Jinjuan Heidi Feng, and Harry Hochheiser. 2017. *Research methods in human-computer interaction*. Morgan Kaufmann.
- [7] Lian Loke, Aaron Blisshen, Carl Gray, and Naseem Ahmadpour. 2021. Safety, Connection and Reflection: Designing with Therapists for Children with Serious Emotional Behaviour Issues. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 162, 17 pages. doi:10.1145/3411764.3445178
- [8] Tom Ongwere, Andrew BL Berry, Clara Caldeira, Rosa I Arriaga, Amid Ayobi, Eleanor R Burgess, Kay Connelly, Patricia Franklin, Andrew D Miller, Aehong Min, et al. 2022. Challenges, tensions, and opportunities in designing ecosystems to support the Management of Complex Health Needs. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts*. 1–7.
- [9] Edward L Schor. 1995. Developing communality: family-centered programs to improve children's health and well-being. *Bulletin of the New York Academy of Medicine* 72, 2 (1995), 413.
- [10] Evropi Stefanidi, Johannes Schöning, Yvonne Rogers, and Jasmin Niess. 2023. Children with ADHD and their Care Ecosystem: Designing Beyond Symptoms. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 558, 17 pages. doi:10.1145/3544548.3581216
- [11] Evropi Stefanidi, Lucas M. Silva, Bengisu Cagiltay, Eva Eriksson, Pawel W. Woźniak, and Jasmin Niess. 2025. *Towards a Research Agenda for Including Children and their Care Ecosystems in HCI*. Association for Computing Machinery, New York, NY, USA, 1225–1227. <https://doi.org/10.1145/3713043.3734465>
- [12] Evropi Stefanidi, Jonathan L.B Wassmann, Pawel W. Woźniak, Gunnar Spellmeyer, Yvonne Rogers, and Jasmin Niess. 2024. MoodGems: Designing for the Well-being of Children with ADHD and their Families at Home. In *Proceedings of the 23rd Annual ACM Interaction Design and Children Conference (IDC '24)*. Association for Computing Machinery, New York, NY, USA, 15 pages. doi:10.1145/3628516.3655795
- [13] Ulrike Zartler. 2010. Multiple perspectives in qualitative family research: Crafting and conducting research projects. *Family Science* 1, 3-4 (2010), 173–182. doi:10.1080/19424620.2010.569371